Robotics 22/01/2021

During the whole exam you should have only one screen active, being connected to the Zoom meeting that you can access from the POLIMI portal.

You should only have a browser open and no other applications. The browser should have only the exam tab open. The exam is saved into the browser cookies so if you close it and reopen it nothing should happen. This is unless you browse in incognito mode or you delete all cookies upon browser closure. Do it at your own risk.

Do not maximize the window of your browser so we can check your desktop too.

When working on paper your camera should look at you working on the paper, when working on the computer the camera should look at you. You cannot go out of sight. Your microphone should be on all the time.

You will have 1.5 hours, after which the Form is automatically closed and there will no possibility to submit your answers anymore. You might be noticed 15' in advance in case you have not uploaded files you should take care of that. Exams that are not submitted within the given time will be considered as RITIRATO.

In case you do not see the IMAGES try to reload the page.

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Punti: 230/270

Kinematics

EXERCISE 1: Robot Kinematics

Consider the robot in the picture.

1

What kind of kinematics does the robot in the picture have? Why?
(10/10 punti)

The robot in the picture has a skid steering type kinematic, this because from the picture it seems that the robot can't steer any of the four wheels, if so it would have been an ackerman type kinematic. If we assume skid steering as the correct type of kinematics of the robot, we can say that the wheels on each side will have the same speed Vl for the left side and Vr for the right side. The robot could also have more than one ICR, caused by the possible slipage of the wheels, but still have the same angular velocity w. Another important parameter that will play a role i the robot kinematic is the baseline B, the baseline of the robot, is the distance between

2

Describe a procedure to compute the exact value of the parameters discussed in the previous answer.
(15/20 punti)

To compute the exact value of the parameters Vl and Vr we could use a sensors that keeps track of the wheels rotations per minute so that, given the wheels radius we could obtain the wheel linear velocity. For the baseline we ca just measure it using a tool.

"You need to perform a calibration procedure for the baseline as the angular velocity is significantly affected by it and an error can significantly propagate. You simply turn around for a given time at constant angular speed. You count the number of revolutions and from that you can estimate the angular velocity and the baseline from that."

3

Write here the kinematics equations relating the wheels' speed to the angular and linear velocities of the robot. Discuss the meaning of the parameters in the equations.
(20/20 punti)

The equations are as follows:
v = (Vr+Vl)/2
w = (Vr-Vl)/2B
R = v/w
Vr is the linear speed of the wheels on the right side and Vl is the linear speed of the wheels on the left side. B is the baseline of the robot, is the distance between the left and right wheel measured in the middle of the wheel. R is the radius of path curvature.

4

Given the previous equations, how would you keep track of the robot position on the plane? Why some formulations are better than others to do that?
(10/20 punti)

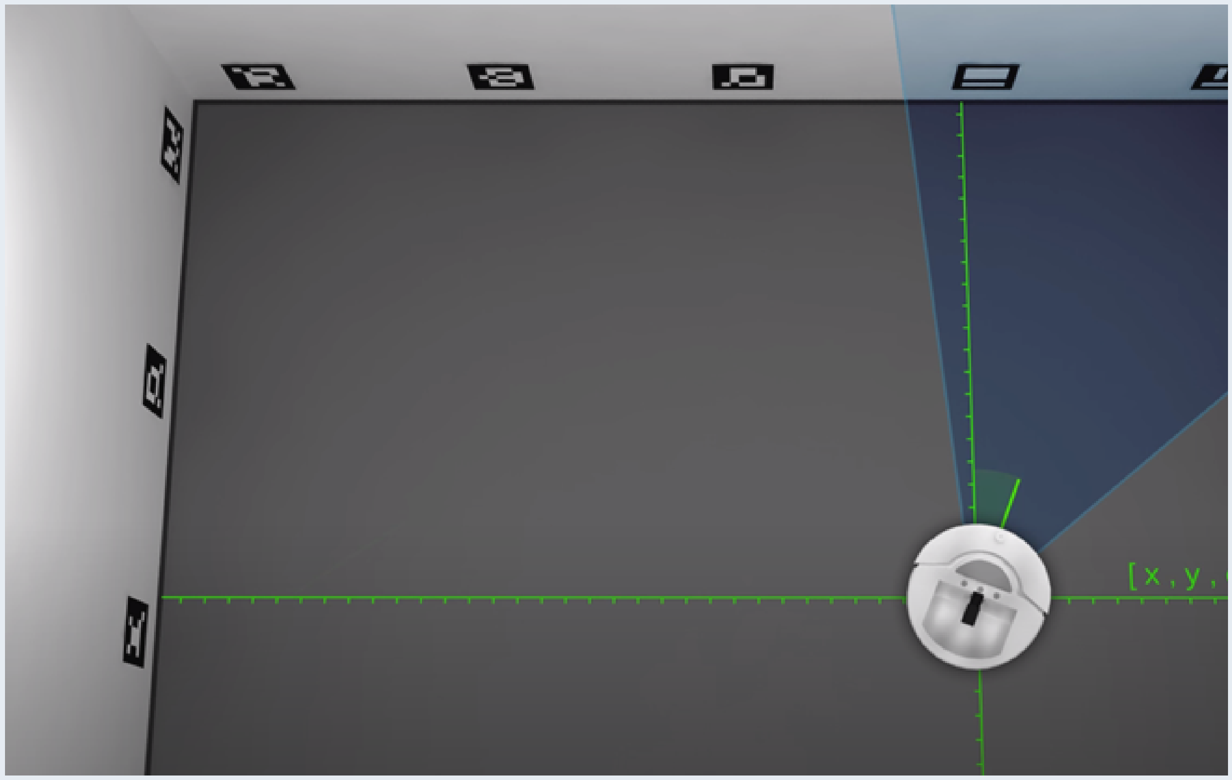
If we know the values of the linear velocities and the baseline , and also know the starting position of the robot, we could keep track of the robot by knowing the period of time the robot has moved at a certain speed and compute the distance travelled and also the final pose. Some formulations are better than others because it makes it easier on the complexity of the

"You were required to answer with the integral of the odometry and discuss the difference between the basic Euler integral and the Rudge-Kutta formulation or the Exact solution ..."

Localization

EXERCISE 2: Localization

In the following, we assume we want to localize a roomba robot in a room instrumented with a set of coded visual markers. Each marker a unique number, however, from time to time, the robot is not able to detect it.



5

Detail how you would encode the map in this application.
(10/10 punti)

I could encode each one of the markers as a landmark on the map, this will make the problem of localization similar to the one of the robo cup seen during lectures. I could also divide the area in cells to create a grid to have a reference of the position of the robot.

6

Write the sensor measurement model in this case according to the description given above.
(15/20 punti)

The sensor measurement model is: P(zt | xt, m), where zt is the measurement, m is the map, encoded with landmarks, and xt is the pose of the robot at the time t.

"It is a range and bearing sensor model, in practice, you should have reported the distance from the marker and the angular direction for the marker. Then you should have also added the fact that the robot might miss the marker from time to time."

7

We want to use an Extended Kalman Filter (EKF) to solve it. How do we implement the motion model?
(0/10 punti)

We can compute the robot odometry using a function x'= g(x, u') which is a gaussian approximation of the motion model of the robot.

"The motion model is a differential drive motion model. In this case g(u(t),x(t-1)) = [x+v*cos(theta)*T, y + v*sin(theta)*T, omega *T], being v and omega the control variables. Then you have to derive to compute the Jacobian."

8

Build the Extended Kalman Filter (EKF) selecting the correct steps (NOTE: Multiple choices apply)
(20/30 punti)

- [ ] mu' = g(mu,u) ✓
- [ ] mu' = G * mu + B * u
- [ ] Sigma' = G * Sigma * G^T + R ✓
- [x] K = Sigma' * H^T * (H * Sigma' * H^T +Q)^-1 ✓
- [ ] K = Sigma' * C^T * (C * Sigma' * C^T +Q)^-1
- [x] mu = mu' + K(z - h(mu')) ✓
- [ ] mu = mu' + K(z - H * mu')
- [x] Sigma = (I - K * H) * Sigma' ✓
- [ ] Sigma = (I - K * G) * Sigma'

9

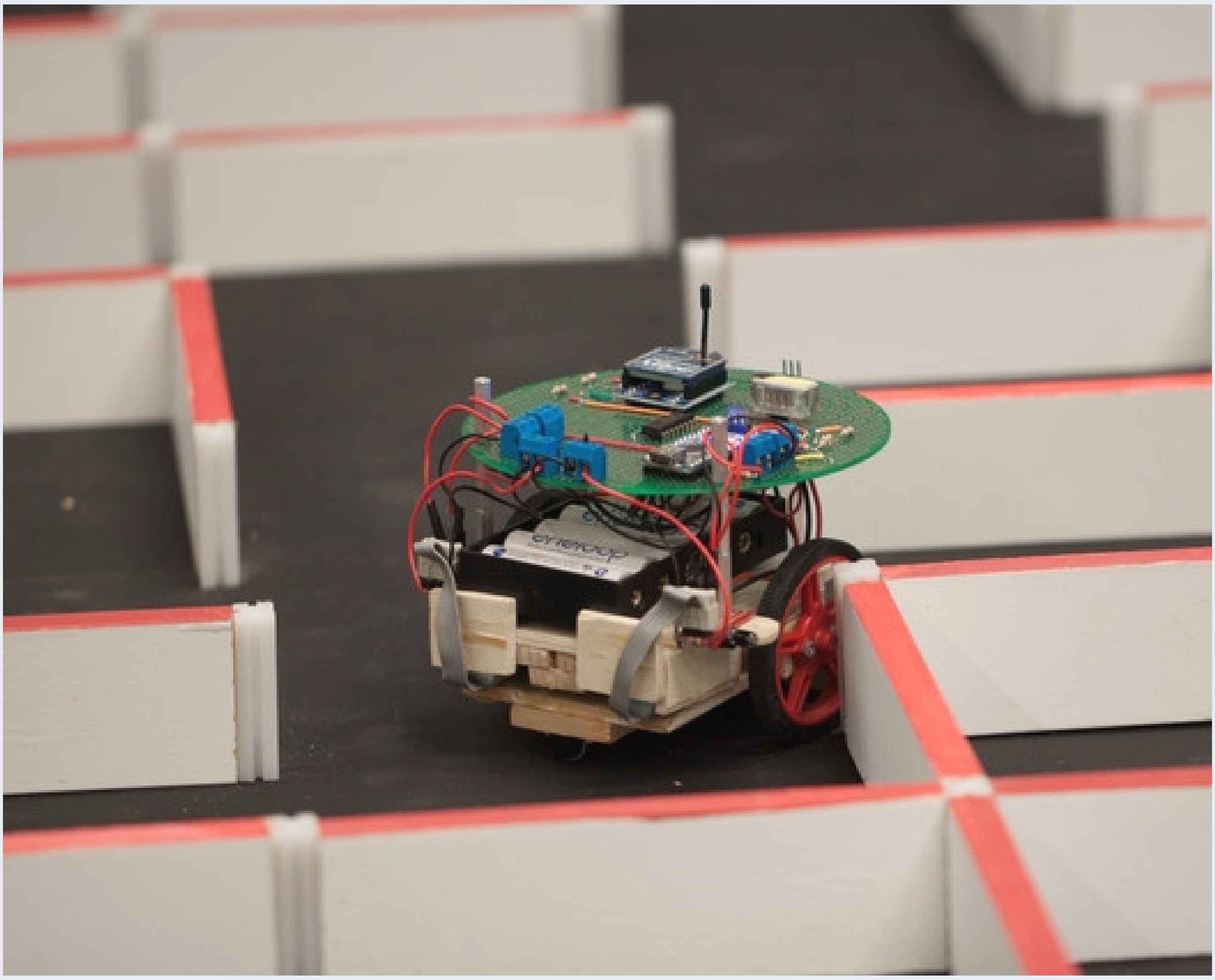
What are H and G in the previous question?
(10/10 punti)

H and G are the jacobian of the h and g functions. The h function is the gaussian approximation of the measurement model while the g function is the gaussian approximation of the motion model of the robot.

Simultaneous Localization and Mapping

EXERCISE 3: Simultaneous Localization and Mapping

Let assume we want to localize in a maze without knowing its layout.



10

Design a SLAM system for this scenario. How would you represent the environment? Which sensors would you use? Which filtering method would you use? Motivate your answers
(30/30 punti)

I would use a 2D map to represent the maze because it the simplest way to represent the map. I would use lasers as sensors so i could use their raytrace function and compute the distances between the walls and the robot. As for filtering methods i would use the particle filter. I wuld do this because this filtering method allows to solve the problem of the kidnapped robot, which is a plus in respect to the other filters and because of his lower computational complexity

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For this problem, it is better to use Online SLAM or a Full SLAM? Why?
(10/10 punti)

We could use the FULL SLAM system because of the fact that this method uses the particle filter and bucause keeps track of the entire robot trajectory.

Motion Control

EXERCIZE 4: Robot Motion Control

Germany has launched the first autonomous Tram! Make it able to leave and stop at tram station autonomously was easy with GPS and vision, but making sure it would have not hit any pedestrian or car crossing the tracks has required to design a motion control system that modulated the speed based on the tracks clearance.

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What are the characteristics of your robot in terms of kinematics?
(10/10 punti)

The robot moves only on tracks, and can't steer his wheels. It can only go forward or backwards. The robot can't rotate in place nor move laterally.

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Which sensors would you chose for the problem of avoiding pedestrians and other vehicles? Why?
(10/10 punti)

I would use lasers as sensor because being in an outdoor envoirment could expose the sensors to high lighting situations when the sun is high and this could disturb other sensors meauseremnt. Lasers use a strong beam of light that makes it more difficult for the sun to disturb measurement. This solution will still be subjected to possible reflections caused by other cars or unexpected objects. I would use a radar ro help with the case of a laser error and also a

14

Which of the obstacle avoidance algorithms would you choose among VHF, VHF+, and DWA for the tram? Why?
(20/20 punti)

I would use VFH+ because it keeps into account for robot kinematics and enlarges obstacles, this is usefoul in a stuation like this one. Also VFH+ checks for arcs trajectories that con occour as the tram moves through the city.

Robot Operating System

EXERCISE 5: Sofware Design

We want to design the obstacle avoidance system for the unmanned tram in the previous exercise. The tram has a radar, a pair of cameras, and a lidar sensor.

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Could we use ROS to implement it? Motivate your answer and, in case it is positive, provide a minimal architecture for it.
(20/20 punti)

We could implement it on ROS. The base functioning of the robot is based on the reading of the measurements of the lidar and the radar. Every time we measure something closer than a certain treshold we immeditely stop the tram. We cold also use another range to define a "slow-zone" where the obstacle isn't too close to the robot and thus we don't need to stop it

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What would you use services for in your system? Make an example and motivate your choice.
(20/20 punti)

I would use servicies to make the robot stop in case of an unexpected obstacle too close to the tram.

EXAM IS OVER!!!

You are one click away from the submission ... double-check you have finished and go!


Are you sur
you
want to do that